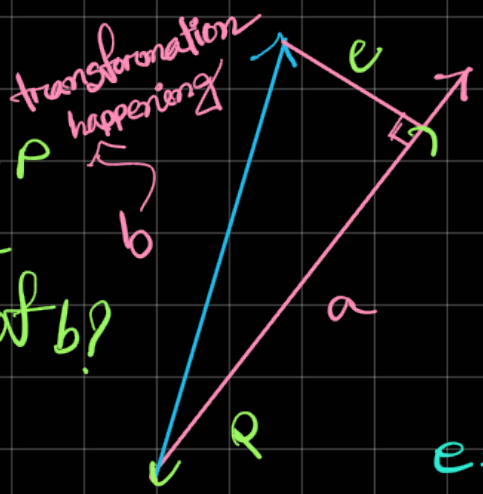
projection





closest point of e from f.

$e = b - xa$

$e = b - p$ —— (I)

$p = xa$ —— (II)

↓ scalar

$e \cdot a = 0$ (perpendicular)

$\Rightarrow a^T e = 0$

$\Rightarrow a^T(b - xa) = 0$

$a^T b - x a^T a = 0$

$\Rightarrow x a^T a = a^T b \Rightarrow x = \boxed{\frac{a^T b}{a^T a}}$ ($a^T b$ = scalar $a \cdot b$)

$Pb = p$ for which transformation

$p = ax$

$= a\left(\frac{a^T b}{a^T a}\right)$

$= \left(\frac{a a^T}{a^T a}\right) b =$ P

$AB \neq BA$

$A(BC) \neq (AB)C$

$[\ ][\ \ \ ]$

$(m\times 1)\ (1\times m) = m\times m$ sq mat

$$\underset{(1\times m)}{\begin{bmatrix} \cdot & \cdot \end{bmatrix}} \underset{(m\times 1)}{\begin{bmatrix} \cdot \\ \cdot \end{bmatrix}} = 1\times 1 \quad a^2$$

$C(p) = Ka$

↓

colspace.

rank of $aa^T \rightarrow 1$

if colspase is a line, we have a single free col.

$$p^T = \left(\frac{aa^T}{a^Ta}\right)^T = \frac{1}{a^Ta}\left((a^T)^T a^T\right)$$

$$= \frac{1}{a^Ta} aa^T$$

$\therefore P = P^T \rightsquigarrow$ symmetric.

Sir's fav property:

$P^2 b = ?$

$\Rightarrow P(Pb)$

$P^2 = P$

$P^n = P \rightarrow$ project P x project P
= same doesn't change.

the length of a doesn't matter only it direction does. ✗

$a.a = a^T a$

$\downarrow$

$a \cdot a \cos 0$

where does the sol^n not exist?

Where does the $\hat{x}$ not exist?

$Ax = b \rightarrow$ not in colspace of A

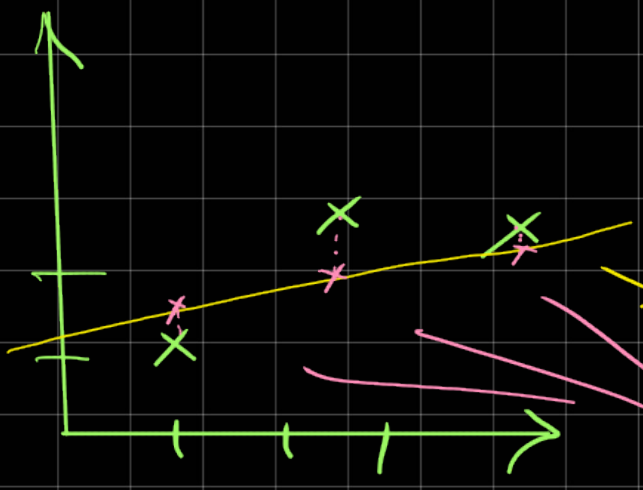
$A\hat{x} = p$

$b \notin C(A)$

↓ not original soln but closest to original like limiting val.

↓ b's proj

$p \in C(A)$



→ No linear soln

→ best fit line.

errors minimized.



Allah save me.

a_1 & a_2 basis vectors that span the colspace

$$p = \hat{x}_1 a_1 + \hat{x}_2 a_2$$

$$e = b - p$$

$\Rightarrow e = b - x_1 a_1 + x_2 a_2$

$$\hookrightarrow \begin{bmatrix} 1 & 1 \\ a_1 & a_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$$

base of Colspace

$\Downarrow$

A

$$\boxed{e = b - A\hat{x}} \quad \text{---} \quad (1)$$

$$a_1^T e = 0 \quad | \quad a_2^T e = 0 \quad | \quad a_1^T (b - A\hat{x}) = 0$$

combine.

$$a_2^T (b - A\hat{x}) = 0$$

row vector

$$\begin{bmatrix} a_1^T \end{bmatrix} (b - A\hat{x}), \quad \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \end{bmatrix} (b - A\hat{x}) = 0$$

$$\begin{bmatrix} a_2^T \end{bmatrix} (b - A\hat{x})$$

$$\begin{bmatrix} 1 & 2 \end{bmatrix}$$

$2x + 4y = 0$ $\rightarrow \begin{bmatrix} 2 & 4 \end{bmatrix}$

$$\begin{bmatrix} -a_1^T- \\ -a_2^T- \end{bmatrix} \square = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boxed{A^T(b - A\hat{x}) = 0}$$

↳ powerful!!

$$A^Tb - A^TA\hat{x} = 0$$

$$A^TA\hat{x} = A^Tb$$

↓

To get only $\hat{x}$,

$Ax = b$

$A^-(Ax) = A^-(b)$

$$(A^TA)^{-1}(A^TA)\hat{x} = (A^TA)^{-1}A^Tb$$

$$\hat{x} = (A^TA)^{-1}A^Tb$$

$A\hat{x} = P$ projection MAT

$$P = A(A^TA)^{-1}A^Tb$$

$= (A (A)^{-1} (A^T)^{-1} A^T)$

$P = b$ whoa! → wrong

if A^{-1} possible we would get I.

but A is not invertible.

Colspace $= A$

(what we multiplied last)

$[A \; AB]$

↓

Col space $= A$

$P^T = P$ (H.W)

$(ABC)^T$

$(A(BC))^T$

$P^2 = P$ proof:

$$A\,(A^TA)^{-1}\,A^T\,\big[A\,(A^TA)^{-1}\,A^T\big]\,b$$

$$= A\,(A^TA)^{-1}\,A^T\,b$$